Narasaraopeta Engineering College (Autonomous)

Department of Computer Science and Engineering

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Batch Number	BB-09
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Guide	Meduri Mounika Naga Bhavani M.Tech
Project Title	**Machine Learning and Deep Learning for Loan Prediction in Banking: Exploring Ensemble Methods and Data Balancing**
Domain/Technology	Deep Learning
Base Paper Link	**https://ieeexplore.ieee.org/document/10772107**
Dataset Link	**https://www.kaggle.com/datasets/subhamjain/loan-prediction-based-on-customer-behavior?utm_source=chatgpt.com**
Software Requirements	Browser: Any latest browser like Chrome Operating System: Windows 11 Server or later Python (COLAB)
Hardware Requirements	System Type: Intel Core i5 or above RAM: 8GB Number of Cores: 5 Number of Threads: 4
Abstract	The increasing volume of loan applications in the banking sector has created a pressing need for efficient systems to predict loan eligibility and defaults. Traditional methods often struggle to handle large datasets and imbalanced class distributions, which can negatively impact predictive accuracy and the reliability of financial decision-making. Several machine learning (ML) algorithms, including Gaussian Naive Bayes, AdaBoost, Gradient Boosting, K-Nearest Neighbors, Decision Trees, Random Forest, and Logistic Regression, were evaluated alongside deep learning (DL) models such as Multi-Layer Perceptron (MLP), Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), Transformer, Gated Recurrent Units (GRU), Autoencoder, ResNet, and DenseNet. To address challenges posed by class imbalance and skewed data distributions, the study also implemented data balancing techniques such as SMOTE and SMOTE-Tomek links, as well as ensemble strategies to further improve predictive performance. The models were assessed using metrics such as accuracy, precision, recall, and F1-score, with DenseNet and ResNet demonstrating superior performance due to their ability to capture complex patterns and nonlinear relationships in the data. Overall, the findings highlight the effectiveness of advanced ML and DL techniques, combined with data balancing and ensemble strategies, in developing reliable and accurate loan prediction systems, thereby supporting financial institutions in optimizing their loan assessment processes, reducing default risks, and making more informed lending decisions.

Signature of the Student(s) **Signature of the Guide** **Signature of the Project Coordinator**